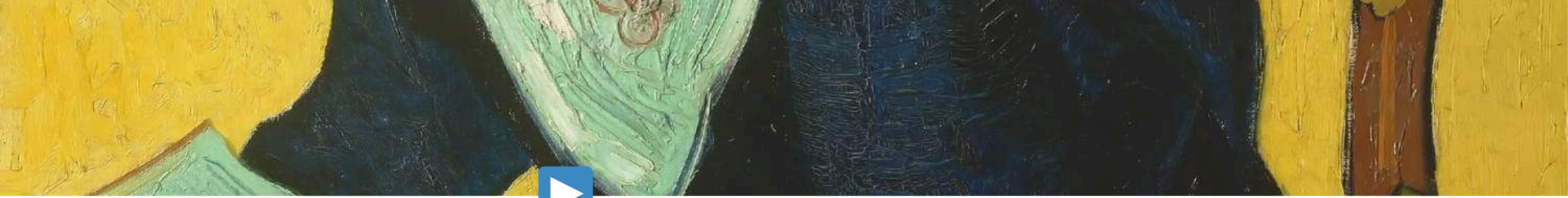


Basics

Use Case: You went to market to buy 1 Chips

- Small Shopkeeper
  - 1 Chips 20 rupees
- Retailer
  - 1 Chips worth 20rupees from Shop1
- Medium Retailer
  - Sahil bought Chips worth 20rupees from Shop1 on 12jan
- Big Retailer
  - Sahil whose phone number is 9682537777 bought lays Chips worth 20rupees from Shop1 served by employee Ravi

Storing Data (in a notebook / text file)

- Sahil whose phone number is 96542537747 bought lays Chips worth 20rupees from Shop1 served by employee Umar on 12 jan
- Aman whose phone number is 9432537747 bought coke worth 30rupees from Shop2 served by employee Ravi on 15jan
- Amit whose phone number is 9684337437 bought lays worth 60rupees from Shop3 served by employee Parth on 18oct
- Shiv whose phone number is 968437437 bought biscuits Chips worth 70rupees from Shop1 served by employee Umar on 21feb
- Swapnil whose phone number is 9682537477 bought lays Chips worth 60rupees from Shop6 served by employee Parth on 13 oct

It is difficult if data is in text file (because searching is difficult)

- Text/Pdf files are made to enter data not to search it

Solution?

- Solution came in 1970
- Save the data in tables

Look at data

- Swapnil whose phone number is 9682537477 bought lays Chips worth 60rupees from Shop6 served by employee Parth on 13 oct

Customers | Products | Shop | Date | Sale

Swapnil | lays | Shop6 | 9oct | 60

Shiv. | Chips | Shop1 | 1 Jan | 70

Swapnil | uncle chipsn | Shop 4 | 12jan | 90

- Called OLTP Data
  - Online Transactional Processing
- All the CRMs, ERPs, generates this kind of data
- Storing data became easy
- Problem?
  - Redundancy
  - Duplicate Data

⇒ Storing data in tables became popular and all the databases came into business

▶ History of Databases

`But there was problem with Databases`

1. Databases were made for writing data into them / Storing data into them. But they were not good in reading the data

- Internal working of database

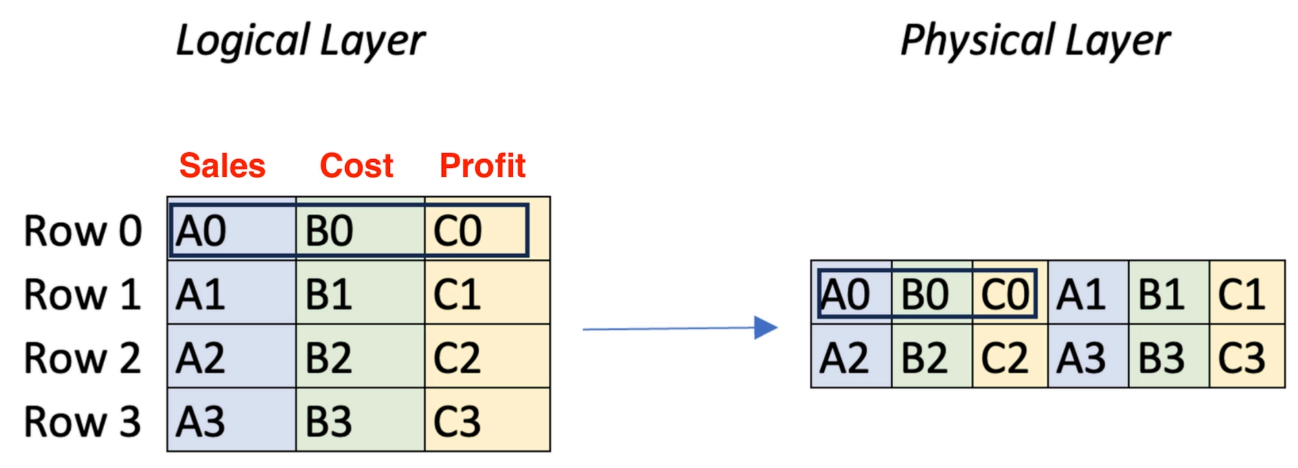


Fig : Row Level Format of Data

To read Total sales from a table of 100Million rows , imagine the slowness

Answer some analytical questions

- What is the total sales of shop?
- How many total profit to shop?

- It's Difficult to use databases (OLTP) in analytics

- To solve this problem, a new concept was introduced called Data warehouses.
  - Data warehouses were made to make reading of data quick and easy due to which we can answer analytical questions easily
  - OLAP Systems
    - Online **Analytical** Processing
  - Called first-generation analytic platforms
  - data warehouses were built with a focus on efficient read operations. The key to their faster and more efficient reads

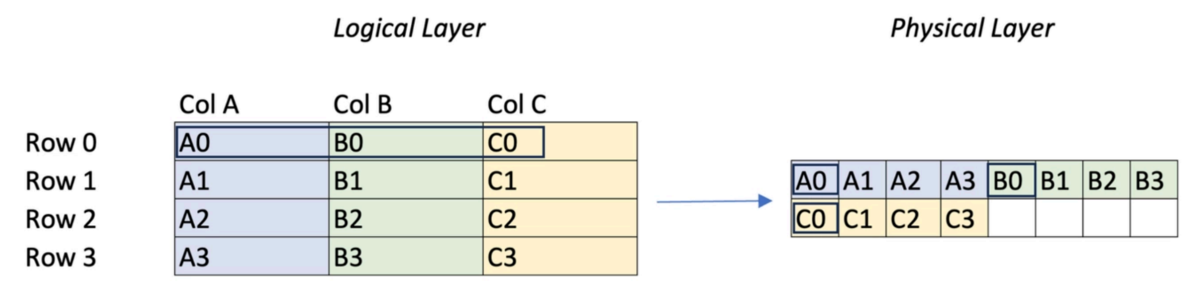


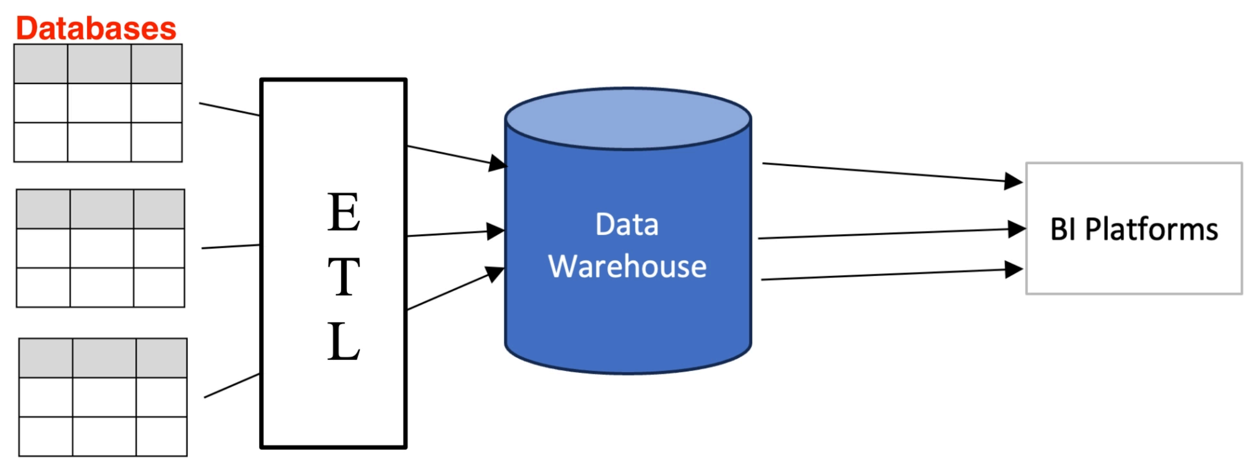
Fig : Column Level format of data

Now, since all the column data is placed side by side, doing aggregation is easy hence analytics is easy

History of Datawarehouses

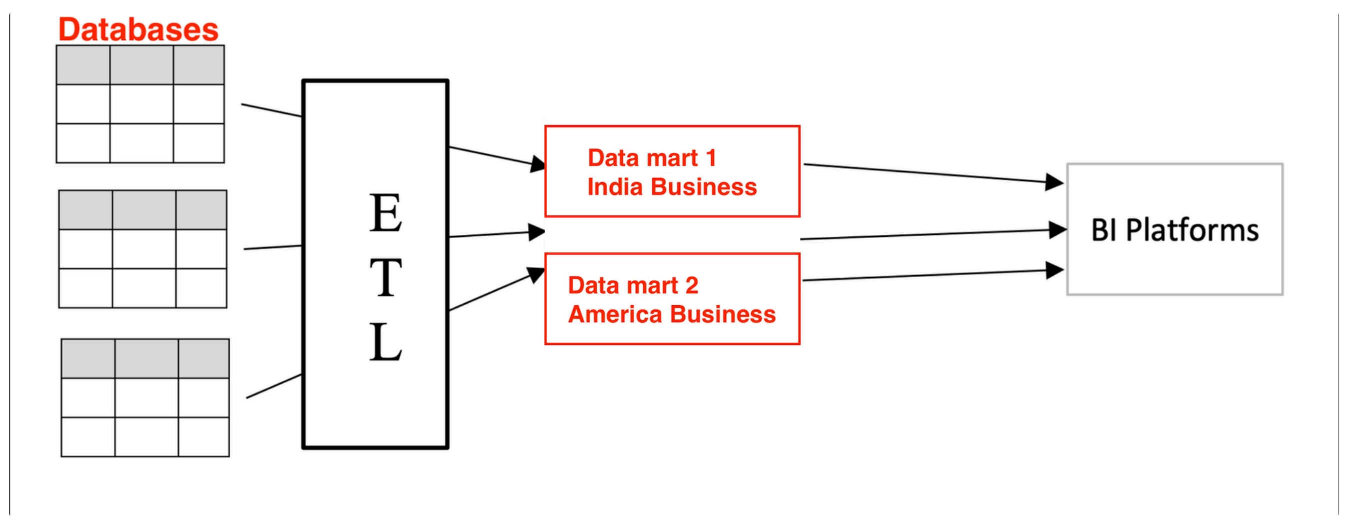
Summary:

- Databases are good to store data
- Datawarehouses are good to read data

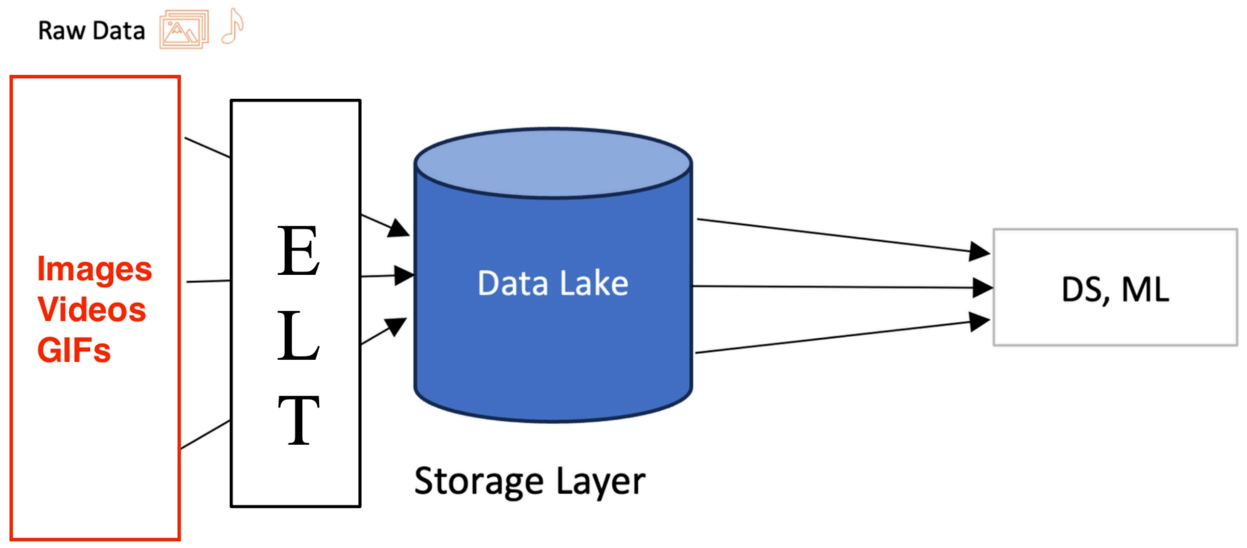


Data architecture of any company looks like this

Concept of Datamarts



Data Lakes


Raw Data
Images
Videos
GIFs
E
L
T
Data Lake
Storage Layer
DS, ML